Conserved coexpression at single cell resolution across primate brains

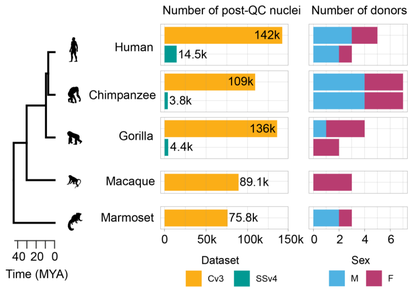
Hamsini Suresh[1], Megan Crow[2], Nikolas Jorstad[3], Trygve Bakken[3], Rebecca Hodge[3], Ed Lein[3], Alexander Dobin[1], Jesse Gillis[1,4,†]
[1]Stanley Institute for Cognitive Genomics, Cold Spring Harbor Laboratory, NY, USA, [2]Genentech, 1 DNA Way, South San Francisco, CA, USA,
[3]Allen Institute for Brain Science, Seattle, WA, USA, [4]Department of Physiology, University of Toronto, ON, Canada

suresh@cshl.edu
jesse.gillis@utoronto.ca

Data: cellular diversity in primate MTG

The Brain Initiative Cell Census Network (BICCN) single-nucleus RNA sequencing datasets sample over 570k nuclei from the middle temporal gyrus (MTG) of five primates (human, chimpanzee, gorilla, macaque and marmoset).

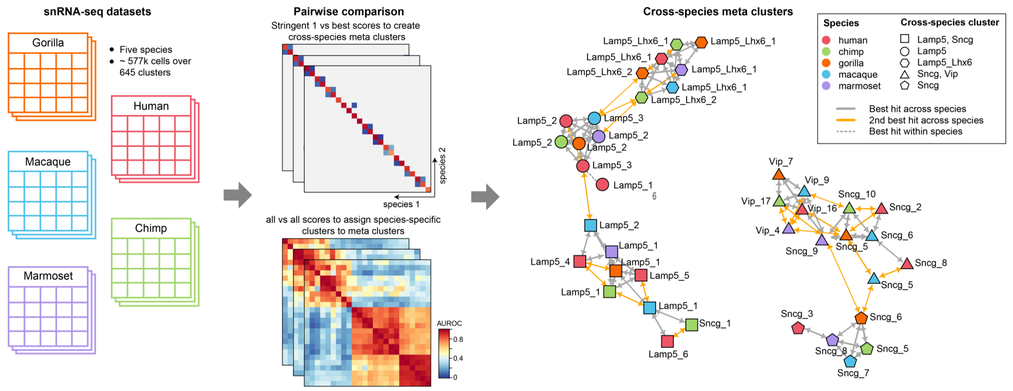
Nuclei were divided into 3 broad classes and 24 subclasses in all species, but clustering at finer resolution revealed different number of cell types across species (112 cell types in marmosets to 170 in humans).



Goal: to explore evolutionary conservation and divergence of MTG cell types and their underlying regulatory mechanisms.

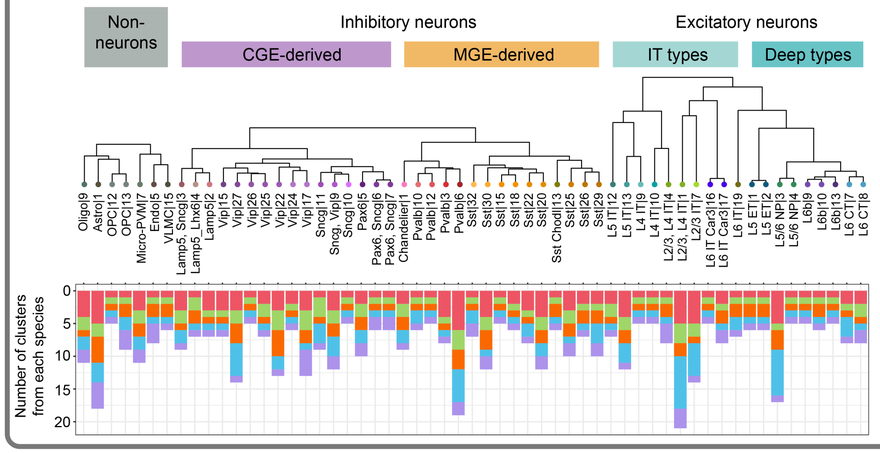
Methods: identifying homologous cell types

We used MetaNeighbor to assess cluster replicability across each species pair, and grouped clusters with high transcriptomic similarity in at least 2 species to create meta-clusters across species.



Consensus MTG taxonomy across primates

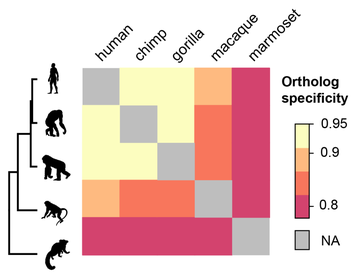
We identified 57 transcriptomically distinct consensus cell types, including 31 inhibitory neuron types, 19 excitatory neuron types, and 7 non-neuronal types.



Hierarchical relationships among cell types roughly mirror their developmental origins.

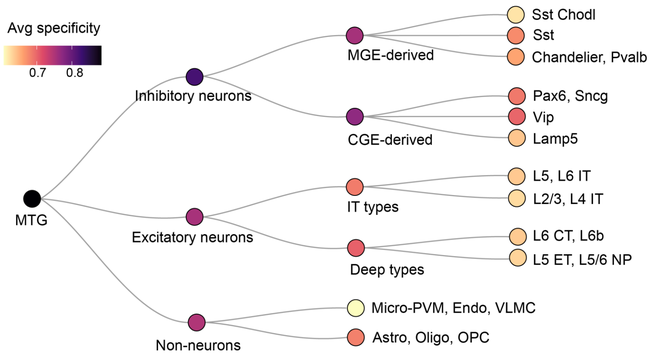
We calculate coexpression similarity over matched cell types for 14k orthologs to measure the extent of functional conservation across species..

Evolutionarily conserved regulatory modules shared across cell types



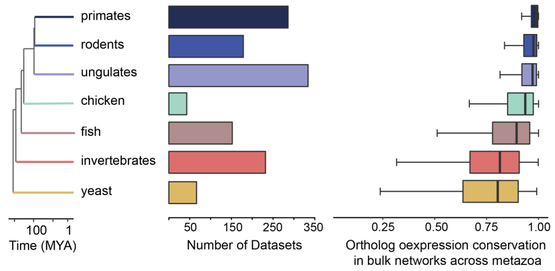
Orthologs have highly conserved expression profiles across the phylogeny (expressologs identified as orthologs with average AUROC = 0.88), with their expression similarities falling with phylogenetic distance.

Stratifying coexpression similarity by cell type hierarchy, we observe consistent coexpression of orthologs at multiple scales (AUROC = 0.83 within meta-clusters and 0.87 within classes), suggesting a core co-regulatory network underlying cell identity across species.

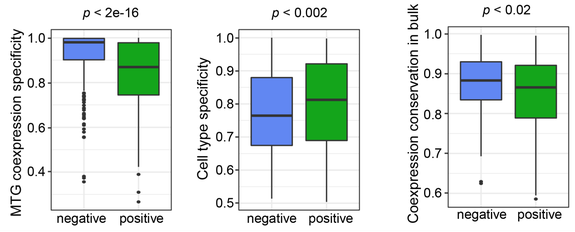


Human-specific regulatory divergence confirmed by integrative analyses with bulk data

Comparison with bulk coexpression networks confirms that genes with conserved expression profiles at single cell resolution also have similar coexpression neighborhoods across metazoa.



Genes under positive selection and genes with changed cellular profiles in humans also exhibit differential coexpression between humans and other species in bulk data, reflecting the changed pattern of activity.



Conclusions

1. Identified 57 transcriptomic cell types across primates with overlapping gene expression patterns.
2. Transcriptional programs that distinguish cell types also exhibit persistent expression variation within cell types defined at finer resolutions.
3. Comparison with large-scale meta-analytic aggregate coexpression networks (derived from 51,978 RNA-seq samples spanning 19 animals) reveals broadly conserved transcriptomic signatures across metazoa along with evidence for divergence.